

# CloudFormation

Class no:26

## What is AWS CloudFormation?

- **Definition:** CloudFormation is an **Infrastructure-as-Code (IaC)** service that lets you **model, provision, and manage AWS resources** using templates.
- **Purpose:** Automate and standardize the creation, updating, and deletion of AWS resources (EC2, S3, RDS, VPC, etc.) in a repeatable, version-controlled way.

## Core concepts

- **Template**
  - A text file in **JSON or YAML** format that describes the AWS resources and their properties.
  - Acts as a **blueprint** for your infrastructure (what to create and how it should be configured).
- **Stack**
  - A **collection of AWS resources** created from a single template.
  - You can **create, update, or delete** a stack, and CloudFormation manages all underlying resources as a single unit.

## Basic template structure (YAML example)

- **AWSTemplateFormatVersion:** Version of template syntax.
- **Parameters:** Inputs you pass when creating/updating the stack (e.g., environment, instance type).
- **Resources:** The actual AWS resources (EC2, S3, etc.) and their properties.
- **Outputs:** Values you can reference later (e.g., bucket URL, instance IP).

## Key features and benefits

- **Infrastructure as Code:** Store templates in **Git / S3**, enabling version control, code reviews, and CI/CD.
- **Repeatable deployments:** Same template always creates the **same infrastructure** in dev, test, and prod (with parameters).
- **Dependency management:** CloudFormation automatically handles **resource ordering and dependencies** (e.g., VPC before subnets).
- **Rollback and change management:** If a stack update fails, CloudFormation can **roll back** to the previous working state. You can use **change sets** to review changes before applying them.

## **Hands on: To create an EC2 template using Cloud formation:**

**Step no:1** Create a **yaml template file** for EC2 and store it as .yaml file.

**Code for reference:** I saved it as **AWSTemplateFormatVersion.yaml**

AWSTemplateFormatVersion: "2010-09-09"

Description: |

Creates a single EC2 instance with a security group allowing SSH.

Parameters:

KeyName:

Type: AWS::EC2::KeyPair::KeyName

Description: Name of an existing EC2 KeyPair to enable SSH access.

InstanceType:

Type: String

Default: t3.micro

AllowedValues:

- t3.micro

- t3.small

Description: Instance type for the EC2 instance.

SSHLocation:

Type: String

Default: 0.0.0.0/0

Description: IP range allowed to SSH to the instance.

Resources:

EC2InstanceSecurityGroup:

Type: AWS::EC2::SecurityGroup

Properties:

GroupDescription: Enable SSH access

SecurityGroupIngress:

- IpProtocol: tcp

FromPort: 22

ToPort: 22

CidrIp: !Ref SSHLocation

EC2Instance:

Type: AWS::EC2::Instance

Properties:

ImageId: ami-0e12ffc2dd465f6e4 # <-- Replace with a valid AMI in your region

InstanceType: !Ref InstanceType

KeyName: !Ref KeyName

SecurityGroupIds:

- !Ref EC2InstanceSecurityGroup

Tags:

- Key: Name

Value: MyCloudFormationEC2Instance

Outputs:

InstanceId:

Description: Instance ID of the EC2 instance

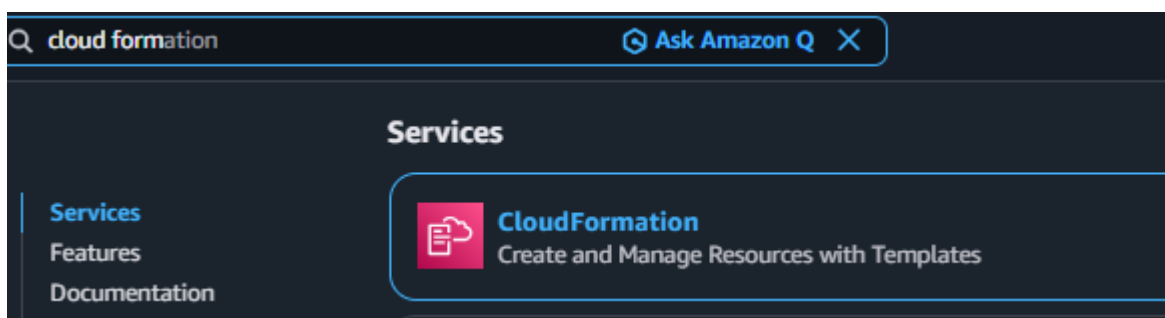
Value: !Ref EC2Instance

PublicIp:

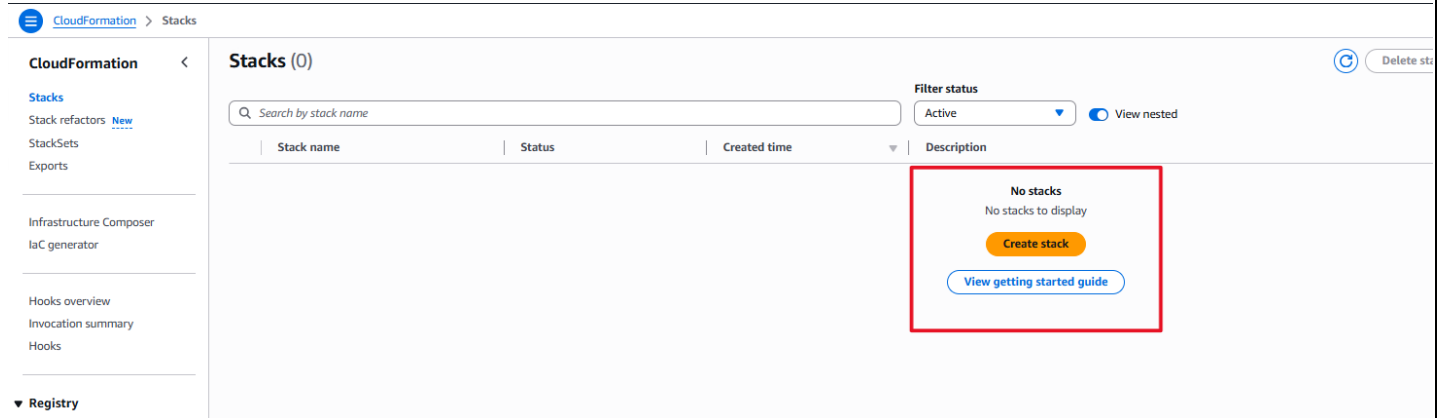
Description: Public IP address of the EC2 instance

Value: !GetAtt EC2Instance.PublicIp

**Step no:2** Go to the AWS console → Search **CloudFormation** → click open.



## Create Stack.



Choose: **Existing template**

### Upload a template file

Choose our file and click **upload**

**Note** the file will be uploaded. AWS will create a bucket and it is saved in it. As we can see the **S3 url** once we have uploaded our file.

### Create stack

#### Prerequisite - Prepare template

You can also create a template by scanning your existing resources in the [laC generator](#).

#### Prepare template

Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

##### ☒ Choose an existing template

Upload or choose an existing template.

##### ☐ Build from Infrastructure Composer

Create a template using a visual builder.

#### Specify template [Info](#)

This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

#### Template source

Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

##### ☐ Amazon S3 URL

Provide an Amazon S3 URL to your template.

##### ☒ Upload a template file

Upload your template directly to the console.

##### ☐ Sync from Git

Sync a template from your Git repository.

#### Upload a template file

##### [Choose file](#)

AWSTemplateFormatVersion.yaml

JSON or YAML formatted file

S3 URL: <https://s3.ap-south-1.amazonaws.com/cf-templates-1wli9mrarvbe-ap-south-1/2026-04-17T120741.681Z2yr-AWSTemplateFormatVersion.yaml>

[View in Infrastructure Composer](#)

[Cancel](#)

[Next](#)

Provide the stack name: **myec2stack**

The **parameters** for the EC2 is read from the yaml file and will be auto populated here.

### Specify stack details

#### Provide a stack name

Stack name

myec2stack

Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 10/128.

#### Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

##### InstanceType

Instance type for the EC2 instance.

t3.micro

##### KeyName

Name of an existing EC2 KeyPair to enable SSH access.

KeyName

##### SSHLocation

IP range allowed to SSH to the instance.

0.0.0.0/0

Cancel

Previous

Next

Keep the rest as default.

### Configure stack options

#### Tags - optional

Tags (key-value pairs) are used to apply metadata to AWS resources, which can help in organizing, identifying, and categorizing those resources. You can add up to 50 unique tags for each stack.

No tags associated with the stack.

Add new tag

You can add 50 more tag(s)

#### Permissions - optional

Specify an existing AWS Identity and Access Management (IAM) service role that CloudFormation can assume.

##### IAM role - optional

Choose the IAM role for CloudFormation to use for all operations performed on the stack.

IAM role name

Sample-role-name

Remove

Create new role



#### Stack failure options

##### Behavior on provisioning failure

Specify the roll back behavior for a stack failure. [Learn more](#)

###### ☒ Roll back all stack resources

Roll back the stack to the last known stable state.

###### ☐ Preserve successfully provisioned resources

Preserves the state of successfully provisioned resources, while rolling back failed resources to the last known stable state. Resources without a last known stable state will be deleted upon the next stack operation.

##### Delete newly created resources during a rollback

Specify whether resources that were created during a failed operation should be deleted regardless of their deletion policy. [Learn more](#)

###### ☒ Use deletion policy

Retains or deletes created resources according to their attached deletion policy.

###### ☐ Delete all newly created resources

Deletes created resources during a rollback regardless of their attached deletion policy.

Next → Next.

Now **Review and submit.**

## Review and create

### Step 1: Specify template

Edit

#### Prerequisite - Prepare template

**Template**  
Template is ready

#### Template

**Template URL**  
<https://s3.ap-south-1.amazonaws.com/cf-templates-1wii9mrarvbe-ap-south-1/2026-04-17T123855.822Zotq-AWSTemplateFormatVersion1.yaml>

**Stack description**  
Creates a single EC2 instance with a security group allowing SSH.

### Step 2: Specify stack details

Edit

#### Provide a stack name

**Stack name**  
myec2stack

#### Parameters (3)

| <input type="text" value="Search parameters"/> |  | < 1 > ⚙   |  |
|------------------------------------------------|--|-----------|--|
| Key                                            |  | Value     |  |
| InstanceType                                   |  | t3.micro  |  |
| KeyName                                        |  | KeyName   |  |
| SSHLocation                                    |  | 0.0.0.0/0 |  |

### Step 3: Configure stack options

Edit

#### Rollback configuration

**Monitoring time**

-

**CloudWatch alarm ARN**

-

#### Notification options

**SNS topic ARN**

No notification options  
There are no notification options defined

#### Stack creation options

**Timeout**

-

**Termination protection**

Deactivated

#### Quick-create link

Use quick-create links to get stacks up and running quickly from the AWS CloudFormation console with the same basic configuration as this stack. Copy the URL on the link to share. [Learn more](#)

[Open quick-create link](#)

[Create change set](#)

[Cancel](#)

[Previous](#)

[Submit](#)

**Step no:3** If there are no issues in the yaml file a stack will be created. The **status become: CREATE\_COMPLETE**.

Stacks (1)

Search by stack name

Filter status: Active View nested

| Stack name | Status          | Created time                 | Description                                                       |
|------------|-----------------|------------------------------|-------------------------------------------------------------------|
| myec2stack | CREATE_COMPLETE | 2026-04-17 18:10:19 UTC+0530 | Creates a single EC2 instance with a security group allowing SSH. |

Go to the **EC2 console** → we can see an instance has been created successfully using the cloud formation.

Instances (1) Info

Last updated less than a minute ago

Saved filter sets: Choose filter set

Find Instance by attribute or tag (case-sensitive)

| Name                        | Instance ID         | Instance state | Instance type |
|-----------------------------|---------------------|----------------|---------------|
| MyCloudFormationEC2Instance | i-037cedbb94c0c73bf | Running        | t3.micro      |

**Step no:4** Now change the name of the instance from **MyCloudFormationEC2Instance** (provided in the template) to **Myinstance**. We can see the drift changes in the cloud formation.

Instances (1/1) Info

Last updated less than a minute ago

Saved filter sets: Choose filter set

Find Instance by attribute or tag (case-sensitive)

| Name       | Instance ID         | Instance state | Instance type |
|------------|---------------------|----------------|---------------|
| Myinstance | i-037cedbb94c0c73bf | Running        | t3.micro      |

Now go to **cloud formation** → **select our stack** → **actions** → **detect drifts**

Stacks (1)

Search by stack name

Filter status: Active View nested

Stack actions: Delete stack Update stack Stack actions Create stack

| Stack name | Status          | Created time                 | Description                                                       |
|------------|-----------------|------------------------------|-------------------------------------------------------------------|
| myec2stack | CREATE_COMPLETE | 2026-04-17 18:10:19 UTC+0530 | Creates a single EC2 instance with a security group allowing SSH. |

- Edit termination protection
- View drift results
- Detect drift
- Create a change set
- Import resources into stack
- Cancel update stack
- Continue update rollback
- View in Application Manager

Click **Detect stack drifts** and refresh.

Drifts

Stack drift status

Drift detection enables you to detect whether a stack's actual configuration differs, or has drifted, from its template configuration. [Learn more](#)

Drift status

⚠️

DRIFTED

Last drift check time

2026-04-17 18:17:41 UTC+0530

🔄

Detect stack drift

📌 This stack contains drifted resources. Create and execute a drift-aware change set to bring all stack resources in line with the template from the previous deployment.

Create a change set

Resource drift status (2)

View drift details

Detect drift for resource

Only resources which currently support drift detection are displayed here. To view all of your stack resources, see your stack details page. [Learn more](#)

🔍 Search resources

< 1 >

| Logical ID               | Physical ID                                                      | Type                    | Drift status | Timestamp                    | Module | Drift status reason |
|--------------------------|------------------------------------------------------------------|-------------------------|--------------|------------------------------|--------|---------------------|
| EC2InstanceSecurityGroup | <a href="#">myec2stack-EC2InstanceSecurityGroup-vbkVUrPdLnIt</a> | AWS::EC2::SecurityGroup | 🟢 IN_SYNC    | 2026-04-17 18:17:42 UTC+0530 | -      | -                   |
| EC2Instance              | <a href="#">i-037cedbb94c0c73bf</a>                              | AWS::EC2::Instance      | ⚠️ MODIFIED  | 2026-04-17 18:17:42 UTC+0530 | -      | -                   |

Now select the **Modified resource** → we can compare the expected template parameters and the actual parameters.

As we changed the name of the instance we can see that here.

Differences (2)

☐

Property

▲

Change

▼

Expected value

▼

Current value

☐

SecurityGroupIds.0

NOT\_EQUAL

myec2stack-EC2InstanceSecurityGroup-vbkVUrPdLnIt

sg-0b84da4f46dcd36a1

☐

Tags.0.Value

NOT\_EQUAL

MyCloudFormationEC2Instance

Myinstance

Details

Expected

Actual

```
{
  "KeyName": "KeyName",
  "ImageId": "ami-0e12ffc2dd465f6e4",
  "InstanceType": "t3.micro",
  "SecurityGroupIds": [
    "myec2stack-EC2InstanceSecurityGroup-vbkVUrPdLnIt"
  ],
  "Tags": [
    {
      "Value": "MyCloudFormationEC2Instance",
      "Key": "Name"
    }
  ]
}
```

```
{
  "ImageId": "ami-0e12ffc2dd465f6e4",
  "KeyName": "KeyName",
  "InstanceType": "t3.micro",
  "SecurityGroupIds": [
    "sg-0b84da4f46dcd36a1"
  ],
  "Tags": [
    {
      "Value": "Myinstance",
      "Key": "Name"
    }
  ]
}
```



## Changing the type of an already existing instance:

**Step no:1** Create and launch an instance.

Once initialized → **instance state** → stop.

Instances (1/1) Info

Connect Instance state Actions

Find Instance by attribute or tag (case-sensitive)

| Name       | Instance ID         | Instance state | Instance type | Status check | Alarm status  | Availability |
|------------|---------------------|----------------|---------------|--------------|---------------|--------------|
| Myinstance | i-037cedbb94c0c73bf | Stopping       | t3.micro      | -            | View alarms + | ap-south-1   |

### i-037cedbb94c0c73bf (Myinstance)

| Details                                                                | Status and alarms | Monitoring                                                                      | Security | Networking                              | Storage | Tags |
|------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------|----------|-----------------------------------------|---------|------|
| ▼ Instance summary Info                                                |                   |                                                                                 |          |                                         |         |      |
| Instance ID<br>i-037cedbb94c0c73bf                                     |                   | Public IPv4 address<br>-                                                        |          | Private IPv4 addresses<br>172.31.46.188 |         |      |
| IPv6 address<br>-                                                      |                   | Instance state<br>Stopped                                                       |          | Public DNS<br>-                         |         |      |
| Hostname type<br>IP name: ip-172-31-46-188.ap-south-1.compute.internal |                   | Private IP DNS name (IPv4 only)<br>ip-172-31-46-188.ap-south-1.compute.internal |          |                                         |         |      |

**Step no:2** Once stopped → click actions → **Instance settings** → change instance type

Instances (1/1) Info

Connect Instance state Actions Launch instance

Find Instance by attribute or tag (case-sensitive)

| Name       | Instance ID         | Instance state | Instance type |
|------------|---------------------|----------------|---------------|
| Myinstance | i-037cedbb94c0c73bf | Stopped        | t3.micro      |

Instance diagnostics

Instance settings

Networking

Security

Image and templates

Storage

Monitor and troubleshoot

Protection and lifecycle management

Attach to Auto Scaling Group

Change stop protection

Change termination protection

Change shutdown behavior

Change auto-recovery behavior

Change instance migration on reboot

Protect with Elastic Disaster Recovery

Instance configuration

Change instance type

Change CPU options

Modify instance placement

Modify Capacity Reservation settings

Change Nitro Enclaves

Connect RDS database

Change credit specification

Change resource based naming options

Edit user data

### i-037cedbb94c0c73bf (Myinstance)

| Details                                                                | Status and alarms | Monitoring                                                                    | Security | Networking | Storage | Tags |
|------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------|----------|------------|---------|------|
| ▼ Instance summary Info                                                |                   |                                                                               |          |            |         |      |
| Instance ID<br>i-037cedbb94c0c73bf                                     |                   | Public IPv4 address<br>-                                                      |          |            |         |      |
| IPv6 address<br>-                                                      |                   | Instance state<br>Stopped                                                     |          |            |         |      |
| Hostname type<br>IP name: ip-172-31-46-188.ap-south-1.compute.internal |                   | Private IP DNS name (IPv4 only)<br>ip-172-31-46-188.ap-south-1.compute.interr |          |            |         |      |

Choose the desired type → Click change instance type.

Change instance type

[Info](#) | [Get advice](#)

You can change the instance type only if the current instance type and the instance type that you want are compatible.

Instance ID

i-037cedbb94c0c73bf

(MyInstance)

Current instance type

t3.micro

New instance type

t3.micro

Change instance type to: t3.micro

c7i-flex.large

m7i-flex.large

t3.micro

t3.small

(32-bit) / (64-bit) architecture can be chosen.

is instance type

|                             | t3.micro            |
|-----------------------------|---------------------|
| On-Demand Linux pricing     | 0.0112 USD per Hour |
| On-Demand Windows pricing   | 0.0204 USD per Hour |
| vCPUs                       | 2 (1 core)          |
| Memory (MiB)                | 1024                |
| Storage (GB)                | –                   |
| Supported root device types | ebs                 |
| Network performance         | Up to 5 Gigabit     |
| Architecture                | x86_64              |
| Hypervisor                  | nitro               |
| Burstable                   | true                |
| Free-tier eligible          | true                |
| Current generation          | true                |

[Compare more instance type attributes](#)

Advanced details

CPU options - update

[Info](#)

☒ Use default CPU options

☐ Specify CPU options

Configure CPUs for your instance to optimize performance and save on licensing costs.

Active vCPUs

2

Total vCPUs

2

Cancel

Change instance type

## **Hands on: To create a S3 bucket using cloud formation**

**Step no:1** Create a yaml file with the S3 configurations and save it as .yaml

### ***Code for reference:***

AWS::CloudFormation::Template

Description: >

Standard Secure S3 Bucket Template.

This template avoids "BucketAlreadyExists" errors by using dynamic naming.

Resources:

# This is the logical ID used inside CloudFormation

MySecureBucket:

Type: AWS::S3::Bucket

Properties:

# NOTE: We omit 'BucketName' so AWS generates a unique one for us.

# This prevents the "Name not available" error you encountered.

# Protects against accidental deletions

VersioningConfiguration:

Status: Enabled

# Encrypts all files uploaded to the bucket

BucketEncryption:

ServerSideEncryptionConfiguration:

- ServerSideEncryptionByDefault:

SSEAlgorithm: AES256

# Best practice: Block all public access by default

PublicAccessBlockConfiguration:

BlockPublicAcls: true

BlockPublicPolicy: true

IgnorePublicAcls: true

RestrictPublicBuckets: true

# Optional: Tags help you organize your billing/resources

Tags:

- Key: Environment

Value: Dev

- Key: CreatedBy

Value: CloudFormation

Outputs:

# This will show you the name AWS chose for your bucket after it's created

GeneratedBucketName:

Description: "The unique name of the bucket created by AWS"

Value: !Ref MySecureBucket

BucketArn:

Description: "The Amazon Resource Name (ARN) of the bucket"

Value: !GetAtt MySecureBucket.Arn

**Step no:2** Now Go to **cloudFormation** → **Create Stack** → **Choose an existing template**

Upload our **S3TemplateFormatVersion1.yaml** file and click next.

## Create stack

### Prerequisite - Prepare template

You can also create a template by scanning your existing resources in the [laC generator](#).

#### Prepare template

Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Choose an existing template

Upload or choose an existing template.

☐

Build from Infrastructure Composer

Create a template using a visual builder.

### Specify template [Info](#)

This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

#### Template source

Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

☐ Amazon S3 URL

Provide an Amazon S3 URL to your template.

☒ Upload a template file

Upload your template directly to the console.

☐ Sync from Git

Sync a template from your Git repository.

#### Upload a template file

[Choose file](#)

S3TemplateFormatVersion1.yaml

JSON or YAML formatted file

S3 URL: <https://s3.ap-south-1.amazonaws.com/cf-templates-1wli9mrarvbe-ap-south-1/2026-04-19T062628.332Zm4p-S3TemplateFormatVersion1.yaml>

[View in Infrastructure Composer](#)

Cancel

Next

Provide a name: **myS3stack**

Specify stack details

Provide a stack name

Stack name

myS3stack

Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 9/128.

Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

No parameters

There are no parameters defined in your template

Cancel

Previous

Next

Review and create.

Review and create

Step 1: Specify template

Edit

Prerequisite - Prepare template

Template

Template is ready

Template

Template URL

https://s3.ap-south-1.amazonaws.com/cf-templates-1wli9mrarvbe-ap-south-1/2026-04-19T062628.332Zm4p-S3TemplateFormatVersion1.yaml

Stack description

S3 Bucket with security best practices.

Step 2: Specify stack details

Edit

Provide a stack name

Stack name

myS3stack

Once submitted, if there are no errors in our template, the S3 bucket creation will be completed.

| Overview                                                                                                                                                                                                                                                                                                                   |                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <div>Stack ID</div> <div> <a href="#">arn:aws:cloudformation:ap-south-1:314383175192:stack/myS3stack/c2578460-3bb9-11f1-9a9c-0270cb6c928b</a> </div> | <div>Description</div> <div>Standard Secure S3 Bucket Template. This te</div>                                                        |
| <div>Status</div> <div> CREATE_COMPLETE</div>                                                                                                                                                                                           | <div>Detailed status</div> <div>-</div>                                                                                              |
| <div>Status reason</div> <div>-</div>                                                                                                                                                                                                                                                                                      | <div>Root stack</div> <div>-</div>                                                                                                   |
| <div>Parent stack</div> <div>-</div>                                                                                                                                                                                                                                                                                       | <div>Created time</div> <div>2026-04-19 12:04:04 UTC+0530</div>                                                                      |
|                                                                                                                                                                                                                                                                                                                            | <div>Updated time</div> <div>-</div>                                                                                                 |
| <div>Deleted time</div> <div>-</div>                                                                                                                                                                                                                                                                                       | <div>Drift status</div> <div> NOT_CHECKED</div> |
| <div>Last drift check time</div> <div>-</div>                                                                                                                                                                                                                                                                              | <div>Termination protection</div> <div>Deactivated</div>                                                                             |
| <div>IAM role</div> <div>-</div>                                                                                                                                                                                                                                                                                           |                                                                                                                                      |

The status becomes: **CREATE\_COMPLETE**

### Stacks (1)

| Stack name                | Status                 | Created time                 |
|---------------------------|------------------------|------------------------------|
| <a href="#">myS3stack</a> | <b>CREATE_COMPLETE</b> | 2026-04-19 12:04:04 UTC+0530 |

We can go to the S3 console → a bucket: **mys3stack-mysecurebucket-ztoo1ubrske** has been created from the template provided in our stack **mys3stack** in cloud formation.

### Buckets

General purpose buckets

All AWS Regions

Directory buckets

#### General purpose buckets (1/1)

Info

Copy ARN Empty Delete Create bucket

< 1 >

| Name                                                  | AWS Region                       | Creation date                        |
|-------------------------------------------------------|----------------------------------|--------------------------------------|
| <a href="#">mys3stack-mysecurebucket-ztoo1ubrs2ke</a> | Asia Pacific (Mumbai) ap-south-1 | April 19, 2026, 12:04:09 (UTC+05:30) |

## **Hands on: To create IAM user, group & policy bucket using cloud formation**

**Step no:1** Create a Template for the IAM user, group and policy using a yaml format and store it in a .yaml file.

### ***Code for reference:***

AWSTemplateFormatVersion: '2010-09-09'

Description: >

Template to create an IAM User with specific S3 permissions.

Resources:

# 1. Create the IAM User

MyApplicationUser:

Type: AWS::IAM::User

Properties:

UserName: !Sub "s3-operator-\${AWS::StackName}"

# Optional: Add the user to a group

Groups:

- !Ref MyUserGroup

# 2. Create an IAM Group (Best practice: manage permissions by groups, not users)

MyUserGroup:

Type: AWS::IAM::Group

Properties:

GroupName: !Sub "\${AWS::StackName}-S3Admins"

# 3. Create a Managed Policy (The "Permissions" document)

S3AccessPolicy:

Type: AWS::IAM::ManagedPolicy

Properties:

Description: "Allows full access to S3 operations"

Groups:

- !Ref MyUserGroup

PolicyDocument:

Version: '2012-10-17'

Statement:

- Effect: Allow

Action:

- "s3:ListAllMyBuckets"

- "s3:GetBucketLocation"

Resource: "arn:aws:s3:::\*"

- Effect: Allow

Action: "s3:\*"

Resource: "\*" # In production, restrict this to specific Bucket ARNs

Outputs:

IAMUserName:

Value: !Ref MyApplicationUser

Description: The name of the IAM user created.

**Step no:2** Now Go to **cloudFormation** → **Create Stack** → **Choose an existing template**

Upload our **IAMTemplateFormatVersion1.yaml** file and click next.

## Create stack

### Prerequisite - Prepare template

You can also create a template by scanning your existing resources in the [JaC generator](#).

#### Prepare template

Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

##### ☒ Choose an existing template

Upload or choose an existing template.

##### ☐ Build from Infrastructure Composer

Create a template using a visual builder.

### Specify template [Info](#)

This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

#### Template source

Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

##### ☐ Amazon S3 URL

Provide an Amazon S3 URL to your template.

##### ☒ Upload a template file

Upload your template directly to the console.

##### ☐ Sync from Git

Sync a template from your Git repository.

#### Upload a template file

##### ☒ Choose file

IAMTemplateFormatVersion1.yaml

JSON or YAML formatted file

S3 URL: <https://s3.ap-south-1.amazonaws.com/cf-templates-1wlii9mrarvbe-ap-south-1/2026-04-19T064401.804Z3i9-IAMTemplateFormatVersion1.yaml>

[View in Infrastructure Composer](#)

[Cancel](#)

[Next](#)



Provide a name: **myIAMstack**

### Specify stack details

#### Provide a stack name

Stack name

myIAMstack

Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 10/128.

#### Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

No parameters

There are no parameters defined in your template

Cancel

Previous

Next

#### ► Stack creation options - optional

Specify the timeout and termination protection options for stack creation.

### Capabilities

① The following resource(s) require capabilities: [AWS::IAM::ManagedPolicy, AWS::IAM::Group]

This template contains Identity and Access Management (IAM) resources. Check that you want to create each of these resources and that they have the minimum required permissions. In addition, they have custom names. Check that the custom names are unique within your AWS account. [Learn more](#)

☒ I acknowledge that AWS CloudFormation might create IAM resources with custom names.

Cancel

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## The "Capabilities" Checkbox

When you upload an IAM template to the CloudFormation console, you **must** check the box at the bottom that says:

**"I acknowledge that AWS CloudFormation might create IAM resources."**

AWS does this because IAM changes can affect the security of your entire account, so it requires explicit consent.

Review and create.

### Review and create

#### Step 1: Specify template

Edit

##### Prerequisite - Prepare template

Template

Template is ready

##### Template

Template URL

<https://s3.ap-south-1.amazonaws.com/cf-templates-1wll9mrarvbe-ap-south-1/2026-04-19T064401.804Z3i9-IAMTemplateFormatVersion1.yaml>

Stack description

Template to create an IAM User with specific S3 permissions.

#### Step 2: Specify stack details

Edit

##### Provide a stack name

Stack name

myIAMstack

Stack creation options

Timeout

-

Termination protection

Deactivated

Quick-create link

Use quick-create links to get stacks up and running quickly from the AWS CloudFormation console with the same basic configuration as this stack. Copy the URL on the link to share. [Learn more](#)

Open quick-create link

Create change set

Cancel

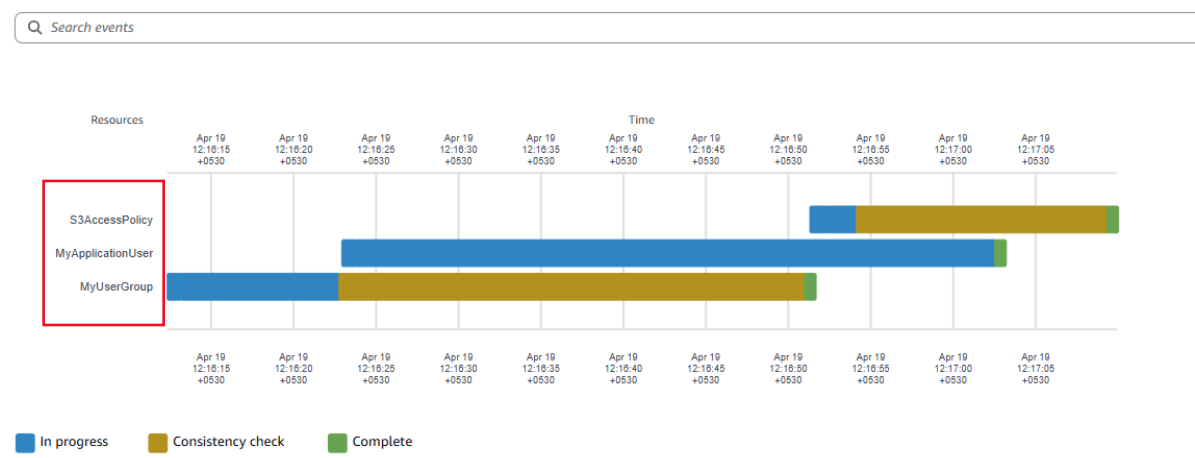
Previous

Submit

Once Submitted, as our template does not have any error we can see the creation for the IAM user , group and policy creation is successful

Latest deployment timeline - new

This is a timeline view of your latest stack deployment. If you start a new deployment, this view will reset.



Events (13)

| Operation ID                         | Timestamp                    | Logical ID        | Status             | Detailed status        | Status reason                        |
|--------------------------------------|------------------------------|-------------------|--------------------|------------------------|--------------------------------------|
| 5d735089-11e6-4984-9a58-2e78b3190558 | -                            | -                 | -                  | -                      | -                                    |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:17:09 UTC+0530 | myIAMstack        | CREATE_COMPLETE    | -                      | -                                    |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:17:09 UTC+0530 | S3AccessPolicy    | CREATE_COMPLETE    | -                      | -                                    |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:17:02 UTC+0530 | MyApplicationUser | CREATE_COMPLETE    | -                      | -                                    |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:54 UTC+0530 | S3AccessPolicy    | CREATE_IN_PROGRESS | CONFIGURATION_COMPLETE | Eventual consistency check initiated |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:52 UTC+0530 | S3AccessPolicy    | CREATE_IN_PROGRESS | -                      | Resource creation Initiated          |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:51 UTC+0530 | S3AccessPolicy    | CREATE_IN_PROGRESS | -                      | -                                    |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:51 UTC+0530 | MyUserGroup       | CREATE_COMPLETE    | -                      | -                                    |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:23 UTC+0530 | MyApplicationUser | CREATE_IN_PROGRESS | -                      | Resource creation Initiated          |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:22 UTC+0530 | MyApplicationUser | CREATE_IN_PROGRESS | -                      | -                                    |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:22 UTC+0530 | MyUserGroup       | CREATE_IN_PROGRESS | CONFIGURATION_COMPLETE | Eventual consistency check initiated |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:13 UTC+0530 | MyUserGroup       | CREATE_IN_PROGRESS | -                      | Resource creation Initiated          |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:12 UTC+0530 | MyUserGroup       | CREATE_IN_PROGRESS | -                      | -                                    |
| 5d735089-11e6-4984-9a58-2e78b3190558 | 2026-04-19 12:16:10 UTC+0530 | myIAMstack        | CREATE_IN_PROGRESS | -                      | User Initiated                       |

The stack status becomes, **CREATE\_COMPLETE**.

| Stack name                 | Status                 | Created time                 | Description                                                  |
|----------------------------|------------------------|------------------------------|--------------------------------------------------------------|
| <a href="#">myIAMstack</a> | <b>CREATE_COMPLETE</b> | 2026-04-19 12:16:10 UTC+0530 | Template to create an IAM User with specific S3 permissions. |

**Step no:3** Now go to IAM → Group → We can see a group: **myIAMstack-S3Admins**

**myIAMstack-S3Admins** Info

Summary

User group name  
myIAMstack-S3Admins

Creation time  
April 19, 2026, 12:16 (UTC+05:30)

ARN  
arn:aws:iam::314383175192:group/myIAMstack-S3Admins

Users (1)

Permissions

Access Advisor

Users in this group (1)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

User name

s3-operator-myIAMstack

Under user: **s3-operator-myIAMstack**

With the policy: **myIAMstack-S3AccessPolicy-pojWpWDrxtxD** for S3 access.

**s3-operator-myIAMstack** Info

Summary

ARN  
arn:aws:iam::314383175192:user/s3-operator-myIAMstack

Console access  
Disabled

Created  
April 19, 2026, 12:16 (UTC+05:30)

Access key 1  
Create access key

Last console sign-in  
-

Permissions

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

Search

Filter by Type  
All types

Policy name

myIAMstack-S3AccessPolicy-pojWpWDrxtxD

Type  
Customer managed

Attached via  
Group myIAMstack-S3Admins

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

It is a customer managed policy created from the details provided in the template.

**Policies (1486)** Info

A policy is an object in AWS that defines permissions.

Search

Filter by Type  
Customer managed

matches

Policy name

Type

Used as

Description

myIAMstack-S3AccessPolicy-pojWpWDrxtxD

Customer managed

Permissions policy (1)

Allows full access to S3 operations